

LARGE PRICE DECLINES, NEWS, LIQUIDITY, AND TRADING STRATEGIES: AN INTRADAY ANALYSIS

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Abstract

This paper examines whether trading strategies based on short-term price reversals following large one-day losses have economically significant returns. We directly incorporate transactions costs by basing returns on the contemporaneous bid and ask quotes and jointly examine the effects of overreaction, liquidity pressure, and public information flow measures. Consistent with the overreaction hypothesis, trading strategy returns increase in the magnitude of event day loss. Consistent with behavioral models, the reversals are higher for event stocks without concurrent news releases. The evidence is generally supportive of the liquidity pressure hypothesis. The analysis suggests refined trading strategies yielding economically significant positive returns. The results are robust to a number of alternative tests.

I. Introduction

Since the publication of De Bondt and Thaler's (1985, 1987) papers showing evidence of reversals in long-term stock price movements, there have been numerous papers examining price reversals over different time horizons and across different markets.¹ This paper addresses the implications of such findings for market efficiency by focusing on whether trading strategies based on short-term price reversals have economically significant returns. Contrary to existing research, such as Atkins and Dyl (1990), Bremer and Sweeney (1991), and Park (1995), this study provides evidence of economically significant short-term price reversals.

To arrive at these results, we directly incorporate transactions costs into the measure of trading strategy returns by using intraday bid and ask quotes. Specifically, we examine a trading strategy in which stocks with large one-day losses during the years 2000 - 2001 are bought at the average of the ask quotes posted during the last 15 minutes of the event day trading session and sold at the bid quotes observed at various points in time during the next trading day. The return measure is subjected to a cross-sectional analysis testing several theories, which have been suggested as explanations of price reversals. Based on the cross-sectional results, simple refinements of the trading strategy have economically and statistically significant returns (e.g.,

¹ Lehman (1990) and Lo and MacKinlay (1988, 1990) show overreaction in U.S. equity markets and suggest a contrarian trading rule that involves buying portfolios of losers and selling portfolios of winners. Jegadeesh (1990) shows profits of approximately 2% per month from following a contrarian strategy whereby stocks are bought and sold based on previous month's returns and held for one month. Lakonishok, Shleifer and Vishny (1994) provide additional evidence of profitability of contrarian/value strategies and address the issue of their intrinsic riskiness. They show that the superior returns from such strategies are not a compensation for extra risk exposure. Along similar lines, La Porta (1996) analyzes whether the profitability of contrarian strategies is caused by systematic errors in expectations. He surveys stock market analysts to test for the existence of systematic errors and provides evidence showing that contrarian strategies based on errors in analysts' forecasts earn superior returns because analysts' expectations about future earnings growth are too extreme. In a follow-up article, La Porta, Lakonishok, Shleifer, and Vishny (1997) examine price reactions to earnings announcements for value and glamour stocks. They suggest that a considerable fraction of return differential between value and glamour stocks that eventually may drive the profitability of value strategies can be attributed to differential earnings surprises, since they are more frequently positive for value stocks.

buying large capitalization stocks with relative losses over 30% and high event day trading volume yields average overnight returns of 1.10%).

Unlike our results, prior studies do not find economically significant returns after indirectly accounting for transaction costs. These studies find statistically significant reversals that do not, in general, represent profitable trading strategies after deducting a typical bid-ask spread.² We expect that using intraday quotes in our analysis will yield a more precise measure of the economic significance of price reversals for several reasons.

First, previous research based on transaction prices does not directly incorporate transaction costs when addressing the economic magnitude of returns from trading strategies based on price reversals.³ It is common to account for transaction costs by subtracting a fixed percentage believed to represent the average spread from the trading rule returns, or to use spreads computed at a point in time removed from the event. This is an ad-hoc adjustment, as transaction costs vary widely with time and security characteristics.⁴ It is likely, for instance, that the bid-ask spread is higher around events that induce increased return volatility (e.g., around negative news releases triggering rapid price declines).

Secondly, given that large close-to-close daily price changes are followed by reversals when one examines daily closing prices rather than intraday data (e.g., see Atkins and Dyl 1990; Bremer and Sweeney 1991), we hypothesize that reversals would materialize at or soon after the beginning of the trading session of the day following the initial price move. Kramer (2001), for

² An exception is Fung, Mok, and Lam (2000) where reversals in the S&P 500 Futures market are examined. The authors show that even after transaction costs profitable trading strategies exist, although their economic significance is marginal.

³ An exception is Akhigbe, Gosnell, and Harikumar (1998) where losing stocks are assumed to be bought at the opening ask and sold at the closing bid. The authors do not examine the overnight and intraday returns, however, which are the primary focus of this study.

⁴ Examples of studies that show evidence of substantial cross-sectional and time series trading cost variability are Keim and Madhavan (1997), Lesmond, Ogden, and Trzcinka (1999) and Lesmond, Schill, and Zhou (2003).

example, finds that essentially all the daily returns are on average realized within the first hour of trading.⁵ Similarly, Harris (1986) shows that the predominant portion of stock price moves takes place within the first 45 minutes of trading. Furthermore, if there is any price adjustment to the previous day's information, the price behavior at the beginning of the trading session is more likely to be a function of the events of the prior day than it is toward the end of the trading session. Thus, one can expect that the cross-sectional variability of reversals will be lower early in the trading session, making them more salient.

Inferences of reversal studies based on transaction prices are also obscured by bid-ask bounce and nonsynchronicity problems, the extent of which becomes increasingly severe as the examination time span shortens. Basing our analysis on quotes eliminates the bid-ask bounce and mitigates the nonsynchronicity problems.

The trading strategy returns are analyzed in cross-sectional regressions based on existing theories that suggest price reversal explanations related to overreaction, liquidity, and public information flow. Besides contributing a comprehensive empirical analysis of these theories to the literature, the cross-sectional analysis is motivated by the following observation: while the average magnitude of price reversals is often relatively small, their cross-sectional variability tends to be quite high. Therefore, if variation around the mean is a function of theoretically motivated characteristics, it is possible that market participants can identify profitable trading rules based on subsets of event firms.

Prior studies frequently suggest overreaction of investors to major news releases as the underlying cause for the subsequent reversals, although little empirical research analyzes the information flow in the context of return reversals explicitly. We directly examine the relevance

⁵ Kramer (2001) finds that the average realized return for the first hour is from 26 to 78 times larger than the average afternoon hour return.

of the news issues by collecting an extensive measure of the public information flow for each event and assessing its effects on the contrarian returns.

In the cross-sectional analysis, we find evidence consistent with the overreaction hypothesis to the extent that trading strategy returns increase in the absolute value of the event day loss. Consistent with models by Daniel, Hirshleifer, and Subrahmanyam (1998) and Hong and Stein (1999), which predict investor underreaction to news and overreaction for extreme price moves unaccompanied by public information releases, we find higher returns for events without concurrent public news releases.

Our evidence is also generally supportive of price reversal explanations based on temporary liquidity pressure, as suggested by Grossman and Miller (1988) and Jegadeesh and Titman (1995) to the extent that returns are found to increase in event day trading volume.

Using the results of the cross-sectional analysis, we arrive at simple refinements of the trading strategy, which yield average overnight returns of between 1% and 2%, if only stocks with capitalization and trading volume in the top sample quartiles are examined. The results are robust to a number of alternative tests.

II. Methodology

Explanations of return reversals

De Bondt and Thaler's (1985, 1987) overreaction hypothesis is based on the psychological phenomenon that individuals tend to assign excessive weight to recent information. Thus, when investors obtain new information, they initially react too strongly, and this overreaction is subsequently corrected causing a return reversal. One of the main predictions of this theory is that since return reversals "correct" previous mistakes, they should

be proportionate to the initial valuation error. In our study, this suggests a positive relation between the absolute value of event day loss and the magnitude of the return based on the price reversal.

While overreaction of investors to new information has often been offered as an explanation for reversals following large stock price moves, there is little existing research that directly relates reversals to the releases of new information.⁶ Larson and Madura (2003) examine whether the over- or underreaction of stocks with daily returns greater than 10% in absolute value is related to concurrent news releases in the *Wall Street Journal*. They examine abnormal daily returns following the events and find evidence of greater overreaction for “uninformed events” – those with no *WSJ* explanation. Using monthly return data and the Dow Jones Interactive Publication Library, Chan (2003) shows that event stocks with news releases tend to exhibit momentum while stocks unaccompanied by public news exhibit reversals.⁷

In our study we further extend this line of research by analyzing overnight and intraday reversals in the context of a new, relatively comprehensive, measure of information arrival compiled from numerous electronic public news sources. Similar to Roll (1988), it is assumed that public information immaterial enough not to be covered by the media is also unimportant in its impact on stock prices.

⁶ Some researchers take the alternate route and deduce the information characteristics from the price changes. For example, see Fabozzi, Ma, Chittenden, and Pace (1995).

⁷ In a related branch of literature, several studies attempt to link stock returns and volatility to real economic events. Roll (1988), for instance, in his examination of how well the price movements of individual stocks can be explained by general economic influences, industry factors, and firm-specific news, finds that after removing all days surrounding firm-specific news releases on the Dow-Jones service, there is only a trivial change in explanatory power as measured by R^2 . Interestingly, Roll (1988) finds several outlier firms for which the explanatory power changes considerably. Such firms tend to face extraordinary news events (e.g., takeovers or mergers). Situations we examine are of similar prominence, given the magnitude of the change in stock price. Most analyses relating aggregate stock returns to aggregate measures of public information find only weak relations. Mitchell and Mulherin (1992) note that since most of the information is firm specific, the relation is obscured by the aggregation process. They devise a measure of firm specific returns and present evidence that it is significantly correlated with public information flow. For more examples of studies analyzing the links between patterns in financial markets and the presence of news reports, see Berry and Howe (1994), Cutler, Poterba, and Summers (1989), Haugen, Talmor and Torous (1991), Ederington and Lee (1993) and Penman (1987).

Daniel, Hirshleifer, and Subrahmanyam (1998) and Hong and Stein (1999) present models that predict investor underreaction to news and overreaction for extreme price moves unaccompanied by public informational releases. Thus, we posit that firms that appear to have no news releases should, all else equal, have a higher likelihood of subsequent reversals. An alternative motivation of this hypothesis is that for firms with news releases, price changes could represent a revaluation effect in light of the new information and should be more permanent compared to firms with no such informational effects.

Jennings and Starks (1986), in their analysis of stock price adjustment to releases of quarterly earnings using samples of companies with and without options listed on their stock, find that firms without options require substantially more time to adjust (up to nine trading hours). Thus, if option markets provide a preferred outlet for informed investors and increase the speed and efficiency with which security prices adjust to new information, then, all else equal, stocks with options listed on them should reverse less, if at all.⁸ Given faster adjustment for stocks with options, it can also be argued that overreaction-driven reversals would be more likely to materialize within the event day.⁹

Peterson (1995) looks at the effect of options trading on stock price adjustment following large daily declines and finds that the three-day cumulative abnormal returns are significantly lower for option firms, suggesting that options improve liquidity and enhance market efficiency. We add to this literature by examining the impact of option listing on reversals for a time span that has not previously been analyzed and while controlling for a number of additional factors potentially related to reversal magnitude.

⁸ Manaster and Rendleman (1982) suggest that informed investors prefer to trade in option markets.

⁹ Jennings and Starks (1986), for example, find that whereas it takes as long as nine hours for non-option stocks to adjust to earnings information, the adjustment of option stocks is remarkably faster – different testing procedures show that it takes anywhere from 15 minutes to two hours.

Grossman and Miller (1988) and Jegadeesh and Titman (1995) show that reversals can result from lack of liquidity in the markets to counter short-term pressures on the buying or selling side. Blume, Mackinlay, and Terker (1989) analyze the return behavior after the October 1987 crash and find that stocks that experienced higher trading volume on the day of the crash also experienced higher subsequent recoveries, suggesting that the selling pressure moved prices down further than warranted and that the returns that followed corrected the preceding declines. Stoll and Whaley (1990) show that prices established on high volume days tend to be reversed at the open of the next trading session, when the inventory imbalances of liquidity providers are liquidated, compensating the latter for the immediacy service. Similarly, Campbell, Grossman, and Wang (1993) find that high volume day returns are likely to revert. Thus, we hypothesize that companies with higher event day trading activity and lower capitalization are likely to have experienced higher liquidity pressure and should reverse more.

Sample selection

All Center for Research in Security Prices (CRSP) securities are sorted by daily close-to-close returns for each trading day of the years 2000 and 2001 and those with losses in excess of 10% on any given day are selected.¹⁰ Data on trading volume, number of trades, and prior trading day capitalization for each firm-day are also taken from CRSP.

We then obtain intraday quotes from the NYSE Trade and Quote (TAQ) database for each company for the event day and the day following it.¹¹ We also require that sample firms have at least one posted ask quote within the last fifteen minutes of trading on the event day.¹² Because for low priced stocks the close-to-close return can exceed the filter of -10% merely due

¹⁰ The filter of 10% was chosen primarily to render our results more comparable to those of prior studies (e.g., Bremer and Sweeney 1991; Cox and Peterson 1994).

¹¹ To minimize the effect of erroneous posts, we disregard those that deviate by more than 40% from the mean daily level.

¹² Firms that do not meet the latter requirement have on average 27 times fewer trades on the event day, 18 times lower volume, 3.8 times fewer outstanding shares, and 26 times lower capitalization compared to the firms that do. The average stock price for these firms is below five dollars even before the event day loss. As we exclude penny stocks from our analysis, this requirement is unlikely to affect our results.

to the bid-ask bounce, this study follows the prior literature in excluding firms whose stock price is equal to or less than five dollars at the end of event day trading.¹³ These filters yield a sample size of 33,284 event-firms for 492 trading days and 4,715 unique tickers representing 630 different 4-digit SIC codes.

We then electronically search CBS.MarketWatch.com and its fifteen news providers for news releases on and prior to event dates for each of our sample firms.¹⁴ The list of news providers contains Reuters, BusinessWire, PR Newswire, Edgar Online, RealTime Headlines, Market Pulse, Associated Press, United Press Intl., Futures World News, New York Times, FT.com, and FT MarketWatch News, among others. Unlike prior studies, we do not include post-event days in our search given the relatively timely nature of our news sources. Conducting the search electronically also allows us to have a substantially larger sample and a much more extensive list of news providers compared to those of prior studies.

For 29,938 of our events we are able to locate the ticker on CBS.MarketWatch.com and create a news dummy variable equal to one if there is at least one news release from the closing hour of the trading day preceding the event day to the closing hour of the day the loss was incurred.¹⁵ Given the speed with which most of our news sources make information available to investors and the sizeable losses incurred on the event days, we believe that most of the news releases would be made within this time window.¹⁶

¹³ Some papers use the filter of ten dollars per share. We repeat the analysis using this alternative hurdle and obtain very similar results.

¹⁴ CBS.MarketWatch.com is chosen for two primary reasons: possibility of search automation and breadth of coverage. The setup of CBS.MarketWatch.com readily allows for search automation since search parameters can be specified directly in the URL address and can thus be varied within the program.

¹⁵ Since we do not actually examine the news release contents, an obvious criticism of our approach is that news can be endogenous. Mitchell and Mulherin (1994) address the news endogeneity issue in their study and find that stories recounting price moves represent less than 1% of the headlines they randomly survey. Such releases only introduce noise to the information flow variable and if controlled for, one would expect to find an even stronger effect.

¹⁶ Berry and Howe (1994), for example, find that the bulk of information is released within trading hours.

Data on option listing are obtained from the Chicago Board Options Exchange (CBOE) as of January 1, 2000 and January 1, 2001 for the events in each respective year, and an option dummy variable equal to one for firms with CBOE options listed on their stock and zero otherwise is created.

Table 1 provides descriptive statistics for our final sample. The sample is skewed in the direction of smaller, less frequently traded and lower priced stocks. An average event day loss is 14% before adjusting for the event day market return and 13% after such an adjustment is made. For approximately 24% of our events, we were able to locate at least one news release and nearly 4 releases on average. In about 61% of the events, the firms had options listed on their stock as of January 1 of the respective event year.

An average event day trade is valued at \$15,171. Barber and Odean (2002), show that individual investors tend to be net buyers of attention grabbing stocks (e.g. those with abnormally high trading volume or a major news release). Consistent with their result, we find that compared to the average dollar value per trade computed over days -250 through -20 , the event day trade size is about 16% lower. The difference is significant at the 0.01 level.

Panel 1 of Figure I shows the monthly distribution of the number of event firms irrespective of the presence of news, and the monthly distribution of event firms with at least one news release. The overall number of firms that have a close-to-close loss of 10% or more varies widely over the sample months. April of 2000 has by far the highest number of event firms at 4,482 - about three times more than the average for the remaining months. One obvious explanation for this variability is the overall performance of the market. In other words, in a month when the whole market declined, one would expect a higher number of firms with daily losses in excess of 10%. Indeed, a regression (not shown) of the daily number of event stocks on

the daily return of the Dow Jones Industrial Index yields a coefficient of negative 29.26 that is highly significant with a t -statistic of negative 4.42.¹⁷ Given this result, we use only the event day loss relative to the return on the CRSP value-weighted index.

Calculation of trading returns

This study assumes that a trader attempting to implement a reversal-based strategy buys stocks that have experienced a large daily loss at the end of the trading session and sells them at various points in time during the next trading day. To take into account the contemporaneous bid-ask spread, we first compute the average of ask quotes posted during the last 15 minutes of trading on the event day for each event firm-day. The average ask is used instead of merely taking the last ask quote to render the strategy more realistic since it is unlikely that a trader can have his buy order(s) executed at the last posted quote. We then subdivide the day following each event day into five-minute increments and obtain 78 bid quotes for every event stock, starting with a quote for 9:35 a.m. through the last quote at 4 p.m.¹⁸ This is done by first allocating all quotes into five-minute time segments and then taking the last quote from each interval. Since quotes are only posted when they are revised, if a quote is missing at any time point the gap is filled by using the previous quote as it still applies.

For each sample firm-day combination the trading strategy return measure is calculated as follows:

¹⁷ It should be mentioned, however, that there are some prominent outliers. The most noteworthy one is April 14, 2000, with by far the highest number of firms exceeding the loss filter of 10% and the highest standardized residual of 7.64. Whether this is related to the tax deadline of April 15 or is a mere happenstance is an interesting question for future research and is not addressed here.

¹⁸ The first quote is taken at 9:35 a.m. as opposed to 9:30 a.m. to make the strategy more realistic and to increase the number of available quotes for the first increment.

$$R_{j,t} = (\text{Bid}_{j,t} - \text{AvgAsk}_j) / \text{AvgAsk}_j \quad (1)$$

Where:

$t = 1, 2, \dots, 78$;

$\text{Bid}_{j,t}$ = bid quote for event j at time increment t ;

AvgAsk_j = the simple average of ask quotes posted during the last 15 minutes of trading on the event day;

If markets are efficient in the sense that if there are any reversals their magnitude is insufficient to exceed the applicable contemporaneous spreads, this return measure will on average be nonpositive.

We use gross unadjusted returns for two reasons. First, given the short investment time spans under consideration, the normal returns are expected to be almost indistinguishable from zero. Second, the unadjusted returns enable us to focus on the realistic profits that can be attained from a reversal-based strategy.¹⁹

III. Empirical evidence

Panel 1 of Figure II and Table 2 summarize the trading results for the overall sample. It appears that a trader buying stocks with relative daily losses in excess of 10% and selling them at 9:35 a.m. the next trading day would suffer losses averaging about 1.5%. The magnitude of such losses increases toward early afternoon and then tapers off, exhibiting an overall U-shaped pattern over the trading day and indicating that there continue to be residual adjustments to the prior trading day's events. Given the findings of prior studies that show evidence of U-shaped

¹⁹ In this sense, our return measure is similar to that used by Akhigbe, Gosnell, and Harikumar (1998). The authors compute trading rule profits as follows: $\text{ReversalReturn} = (\text{CloseBid} - \text{OpenAsk}) / \text{OpenAsk}$.

patterns in intraday spreads, and since the return measure we use is an inverse function of the spread, absence of any such residual effects would lead to an inverse U-shaped intraday pattern for the trading returns.²⁰

The evidence of such residual adjustment is consistent with the results of Patell and Wolfson (1984), who examine the extent to which the arrival of dividend and earnings information interrupts the usual reversal and continuation frequencies of intraday prices and the speed with which they return to normal levels. Although the authors show that the announcement effects largely dissipate within one hour to ninety minutes, they find statistically significant departures that continue into the next day. The authors suggest that the evening following the announcement day enables investors who could not execute intraday strategies to receive the information, and their actions then influence the overnight price changes and the opening trades of the next day.

To the extent that specialists might moderate overnight price behavior due to continuity requirements (e.g., see Miller 1989), the reaction will be less noticeable at the open compared to the first minutes thereafter, consistent with the precipitous declines evident in the first minutes of trading seen in Panel 1 of Figure II.²¹

A key result that can be seen both in Table 2, where event firms sharing the same event date are combined into portfolios with weights determined by event day relative losses and in Figure II, where each event is treated independently, is that consistent with the tenets of the overreaction hypothesis, the trading returns appear to be an increasing function of the absolute

²⁰ See Admati and Pfleiderer (1988) for an example of a model that explains the causes for the U-shaped intraday spread pattern.

²¹ Amihud and Mendelson (1990) provide evidence contrary to Miller's findings.

value of the relative loss incurred on the event day.²² As the magnitude of the loss relative to the CRSP value-weighted index increases, the overnight trading returns tend to also rise, although the relation is less conclusive for longer holding periods.

Analysis of information flow

Before presenting the results of the cross-sectional analysis of trading returns, we first discuss the characteristics of the information flow as measured by the presence of firm specific news releases. Because of the uniqueness of the news data, we include several descriptive graphs.

Panel 2 of Figure I shows the variation in the proportion of event firms with news releases across the months. Berry and Howe (1994) and Mitchell and Mulherin (1992) find that November and December are the lightest information months and May and July are the heaviest. They also show that January, April, July, and October have more information because of quarterly reports. To the extent that our information flow measure is conditioned on large daily losses, our results are not directly comparable to theirs.²³

Prior overreaction and reversal studies assume that significant daily losses are caused by the arrival of new information. Panel 2 of Figure I shows that the share of firms with news releases is relatively small when the overall sample is examined. The low incidence of news is a somewhat surprising finding given the sizeable losses incurred, indicating a potential weakness of inferring information characteristics from price changes (e.g. Fabozzi, Ma, Chittenden, and

²² The daily portfolios are created to avoid the potential test bias that can result if the returns on same-day firms are not independent. We thank an anonymous referee for pointing this out.

²³ The share of firms having news releases tends to be higher during the second half of the year. This result could be due to the general propensity of firms to delay conveying bad news until later in the year. Telephone conversations with CBS.MARKETWATCH.COM representatives indicate that the shift cannot be attributed to changes in news coverage.

Pace, 1995). On the other hand, the graph presents an intuitively appealing result in that the proportion of event firms with news is increasing in the absolute value of the relative event day loss. For instance, in September of 2000, we are able to find at least one news release for all of the firms with a relative loss in excess of 30%. The numbers are similar to those in Ryan and Taffler (2002) who show that more than 65% of price and volume movements in the extreme tails of the respective distributions are explained by publicly available news releases.

Berry and Howe (1994) also find that weekends are light information days, and that Mondays and Fridays are light compared to other weekdays, especially Tuesdays and Thursdays. Considering that we combine the news releases made public from 4 p.m. on Friday to 4 p.m. on Monday, our results (see Figure III, Panels 1 and 2) are generally consistent with theirs and inconsistent with the findings of Patell and Wolfson (1982), who show that firms tend to release bad news after the close of trading on Fridays.²⁴ We are able to reject the hypothesis of equal means across weekdays with a p -value of less than 0.0001.

Nofsinger (2001) shows that the number of firm-specific news releases is an increasing function of size. Clearly, larger firms tend to get higher news coverage. The probability of a news release is also potentially related to event day loss and trading activity. We approach this question by estimating several logistic models of the form:

$$\text{News}_j = \beta_0 + \beta_1 \text{RelativeLoss}_j + \beta_2 \text{LogVol}_j + \beta_3 \text{LogCap}_j + e_j \quad (2)$$

Where:

News_j = one if we locate at least one news release for event j from 4 p.m. of the event day to 4 p.m. of the preceding trading day and zero otherwise;

²⁴ Along similar lines, Penman (1987) shows that more bad earnings news arrives on Mondays and (to a lesser extent) on Fridays; Nofsinger (2001) finds that the highest number of firm specific news articles is on Friday.

RelativeLoss_j = absolute value of the difference between the event day close-to-close loss incurred by the firm and the respective return on the CRSP value-weighted index;

LogVol_j = the natural logarithm of event day trading volume;

LogCap_j = the natural logarithm of pre-event day capitalization;

e_j = error term.

Table 3 presents the results of the logistic regressions. We find that companies with a greater event day relative loss and higher event day trading volume are more likely to have a news release. The evidence of a capitalization effect is weaker. Higher capitalization tends to increase the likelihood of a news release, although the relation becomes insignificant when volume is added due to a collinearity issue. The results are generally consistent with the findings of Chan (2003), who shows that the cross-sectional correlations of log market value and turnover with log news citations per month average 0.37 and 0.16, respectively.

Cross-sectional analysis of trading returns

We use cross-sectional analysis to test the theories that have been offered as explanations of price reversals. Running ordinary least squares on the pooled sample can lead to erroneous inferences due to potential error correlations for the event firms that share the same calendar day. Therefore, to control for the day effects we use random effects in our cross-sectional analysis and estimate models of the following form:²⁵

²⁵ OLS and fixed effects lead to similar results. Random effects results are presented based on the Hausman specification test. Several prior studies avoid the correlation problem by alphabetically ranking all event stocks each day and only taking the first firm. Unlike these studies (typically based on daily CRSP returns over several years), we examine intraday data and are limited to only two years due to data constraints. Replicating the analysis with only one event firm per day strongly reduces the sample size and statistical significance, although the directional inferences remain largely unchanged.

$$R_{j,t} = \gamma_0 + \gamma_1 \text{Spread}_{j,t} + \gamma_2 \text{RelLoss}_{j,t} + \gamma_3 \text{LogVol}_{j,t} + \gamma_4 \text{LogCap}_{j,t} + \gamma_5 \text{News}_{j,t} + \gamma_6 \text{N_News}_{j,t} + \gamma_7 \text{Option}_{j,t} + \gamma_8 \text{TradeSize}_{j,t} + v_t + e_{j,t} \quad (3)$$

Where:

$j = 1, 2, \dots, K$; K = number of events;

$t = 1, 2, \dots, T$; T = number of event days;

$R_{j,t}$ = returns from buying at the average of ask quotes posted within the last 15 minutes of trading and selling at the bid quotes at 9:35 a.m. the next trading day;

$\text{Spread}_{j,t}$ = difference between average ask and average bid quotes posted from 3:45 p.m. to 4 p.m. during the event day relative to the midquote point;

$\text{RelativeLoss}_{j,t}$ = absolute value of the difference between the event day close-to-close loss and the return on the CRSP value-weighted index;

$\text{LogVol}_{j,t}$ = natural logarithm of event day trading volume;

$\text{LogCap}_{j,t}$ = natural logarithm of pre-event day capitalization;

$\text{News}_{j,t}$ = one for stocks with news release(s), and zero otherwise;

$\text{N_News}_{j,t}$ = number of news releases;

$\text{Option}_{j,t}$ = one if the stock has a CBOE-listed option and zero otherwise;

$\text{TradeSize}_{j,t}$ = average event day value of a trade;

$v_t, e_{j,t}$ = random error terms.²⁶

Table 4 summarizes the main results. Consistent with the prediction of the overreaction hypothesis, the return from the trading strategy is positively related to the absolute value of the

²⁶ We repeat the analysis using percentile indices instead of natural logarithms for the skewed variables (capitalization and volume) and obtain qualitatively similar results. We also repeat the analysis for longer holding periods up through increment 78. The results are again largely unchanged.

event day loss magnitude. The loss variable coefficient has the predicted positive sign and is economically and statistically significant in all specifications.

The news dummy coefficient is negative and significant in all specifications. Furthermore, there is also weak evidence that the returns are decreasing in the number of news releases. This finding is consistent with the results of Larson and Madura (2003) and Chan (2003). It also yields support to the behavioral models of Daniel, Hirshleifer, and Subrahmanyam (1998) and Hong and Stein (1999). The former predict that investors overreact to private signals; the latter show that investors overreact to price shocks unrelated to the information flow. It is impossible to distinguish between these predictions without more direct data on private information flow.

Nofsinger (2001) shows that the overall news visibility is only significant for small investors and that firm specific news releases do not explain the trading of institutional investors well. We repeat the analysis (results not reported here) for subsets of the sample based on capitalization quartiles and find that, consistent with Nofsinger (2001), the news dummy coefficient is more significant, both economically and statistically, for lower capitalization quartiles.

Unlike Peterson (1995), we do not find that the availability of listed options mitigates return reversals. On the contrary, it appears that option stocks tend to have higher reversals, possibly indicating that non-option (mostly small capitalization) stocks may take longer to correct excessive moves and our examination window is not long enough to show it.

The cross-sectional results support the liquidity pressure hypothesis with respect to measures of trading activity. The coefficient of the trading volume variable is highly significant and has the expected positive sign. On the other hand, contrary to the predictions of the liquidity

pressure hypothesis, the capitalization variable loads positively and is significant in all but one specification. This finding is consistent with the results of Larson and Madura (2003), who, using a longer window of analysis, also find greater overreaction for larger firms.

The effect of capitalization is puzzling. Since the trading strategy return is directly based on the contemporaneous spread and is an inverse function of it, and as spreads are generally lower for large capitalization companies, it is possible that the capitalization variable proxies for the influence of the spread.²⁷ To control for this possibility, we include a percentage spread variable calculated as the difference between the averages of ask and bid quotes posted within the last 15 minutes of event day trading divided by the midquote point. The capitalization variable still loads positively and remains significant and the effects of other variables remain largely unchanged.

Nofsinger (2001) and Blume and Friend (2002) find that institutional investors tend to trade in stocks of large firms whereas individual investors mostly trade in small firm stocks. If it takes longer for individual investors to price the implications of new information, it is possible that we are unable to capture reversals for small companies within our window of analysis. Furthermore, Akhigbe, Gosnell, and Harikumar (1998) suggest that large price changes for small “neglected” firms might attract attention and induce other investors to take positions. This behavior can create short-term momentum for small capitalization stocks.²⁸ Barber and Odean (2002) also suggest that the tendency of small investors to buy stocks with extreme negative returns may contribute to momentum in small capitalization losers. Contrary to these arguments, however, we find that the reversals are a decreasing function of the average event day trade size.

²⁷ Lehman (1990), for instance, suggests that small firms contribute primarily to transactions costs and not to portfolio profits.

²⁸ Hong, Lim, and Stein (2000) test the gradual information diffusion model and show that firm-specific information, particularly of a negative nature, disseminates slowly, giving rise to momentum. The effect is especially strong for smaller firms with lower analyst coverage.

Trading strategy refinements

The theoretical explanations of reversals suggest several ways to refine the trading strategy. We examine three relatively simple refinements whereby the sample is subdivided based on capitalization, trading volume and relative event day loss magnitude. Table 5 summarizes the main results.

Average returns for stocks in the top volume quartile (Panel 6) substantially exceed those for stocks in the bottom volume quartile (Panel 3), and appear to increase in the absolute value of the relative event day loss. Similarly, average returns for stocks with capitalization in the top quartile (Panel 8) substantially exceed those for stocks in the bottom quartile (Panel 7), increase in the absolute value of relative event day loss, and for losses in excess of 30% and 35% equal 0.96% and 1.73% overnight, respectively. Both numbers are significant at the 0.01 level.

Combining the two splits (Panel 5) further enhances the performance of the strategy. A trader focusing only on stocks with capitalization and event day trading volume in the top quartiles and with relative event day losses in excess of 30% or 35% achieves overnight portfolio returns of 1.10% and 1.73%, respectively.²⁹ Furthermore, this strategy offers an additional benefit of ensuring that trades are more promptly executed.

Panel 2 of Figure II shows the average trading strategy returns for stocks with capitalization and trading volume above the respective 75th percentiles over the 78 holding period increments that we examine. The plot is similar to that of the overall sample presented in Panel 1, except that the returns are all generally shifted upwards and the positive effect of the relative loss is more distinct. More importantly, unlike the overall sample results, the returns of the most profitable strategy remain positive throughout the day and are statistically significant

²⁹ For all estimates in Table 2 that have significant *t*-statistics, sign tests and signed rank tests generate even lower *p*-values.

across almost all of increments in the first half of the day. As more time passes and new information potentially unrelated to prior trading day's events arrives, the variability of returns increases and their statistical salience declines.

Figure IV shows the histograms of overnight trading returns for the subset of firms with capitalization and event day trading volume in the top quartiles, and with relative event day losses in excess of 30%. Panel 1 treats each event firm separately, whereas Panel 2 shows the distribution for a realistic strategy in which a portfolio of event stocks is formed each day with weights determined by relative event day losses.

Both the means and the medians are positive and the majority of return realizations are nonnegative. It is, of course, still possible that given several consecutive negative outcomes, the investor's position can considerably decline or be depleted. An examination of the realized returns, however, indicates that negative outcomes are not clustered in time and a dollar invested at the beginning of the sample period with the gross proceeds continually reinvested into new event portfolios each sample day grew to \$2.38 for the case of the strategy examined above, yielding an annual return of 54.29%.

We further assess the reliability of this estimate by conducting a bootstrap procedure. Since there are more event days in 2000 than in 2001 (52 vs. 35) for the above-mentioned strategy, we first randomly determine how many event days the simulated year will have out of these two alternatives and then sample with replacement from the pool of daily trading returns and compute simulated annual returns. The procedure is repeated one million times and the results are reported in Figure V. The mean annual bootstrapped return equals 61.13% and only 5.3% of the outcomes are negative. The minimum and maximum possible returns are -47.70%

and 586.36%, respectively. These results appear to indicate that the original return estimate is not a low-probability outcome of an unusually lucky sequence of daily trading returns.³⁰

To examine whether there is industry clustering among same-day event firms, we perform the following bootstrap procedure. For each event day we randomly draw the firm SIC codes from the empirical distribution and compute the Herfindall-Hirschman Index (HHI). After calculating the concentration index for each sample day, an average HHI is computed for the simulated year. The steps are repeated one million times. In unreported results, we find that the difference between the actual and the simulated HHI's is not statistically different from zero at the conventional levels.

Since a greater proportion of actual trades take place within the quoted spread for larger capitalization stocks, the trading returns computed by our measure are likely to be underestimated. Furthermore, when securities experience large price declines and market makers are rebalancing their inventories, this often enables traders following reversal strategies to trade on more favorable terms since they provide liquidity.³¹ Lehman (1990) gives evidence on practitioners' experience showing that one-way transactions costs, including the price pressure cost, are less than 0.2% on short-term reversal strategies. The median spread for the above-described strategy is 0.56%.

IV. Robustness

Because we arrive at our initial sample by looking at the event day close-to-close loss and then go on to assume that a trader following a reversal-based strategy would attempt to purchase

³⁰ We also conduct a three-step bootstrap estimation procedure: first, we draw the number of event firms for each simulated event day from the empirical distribution; then we draw event firms and calculate 1,000,000 simulated event day returns. In the final step, we draw from the simulated event day returns to obtain 1,000,000 annual returns. The results (available on request) are consistent with the findings of the reported simulation.

³¹ See Lehman (1990) for a discussion of both issues.

stocks thus identified before the market closes, an obvious criticism is that we may have inadvertently included in our sample stocks that only became “identifiable” within the last 15 minutes. Similarly, we may have inadvertently excluded some stocks that attained the filter of 10% at 3:45 p.m. and then went on to increase in value.

We address the former concern by including only stocks that lost 10% or more as of 3:45 p.m. on the event day. The results (not reported here) remain almost identical. To mitigate the possible biases created by the latter issue, we only include stocks in our sample that had lost 20% or more as of 3:45 p.m. on the event day. Arguably, it is unlikely that stocks with losses in excess of 20% would reverse by enough during the last 15 minutes of trading to be excluded from our initial sample. Thus, we believe that this procedure yields a pool of companies very similar to the one we would have had if we had not conditioned the original sample on daily close-to-close losses of 10% or more. Generally, the results (not shown) remain qualitatively and quantitatively unchanged.

In the preceding sections, we assume that the stocks are bought at the average of ask quotes posted during the last 15 minutes of event day trading. As an additional robustness check, we recalculate the returns assuming that the trader’s buy orders are always executed at the highest ask quote posted over this interval. The returns for the most profitable strategies (not reported) remain positive, although they are considerably smaller in magnitude and are not significantly different from zero.³²

³² As a nonparametric check of the cross-sectional results, we also conduct cluster analysis for the subset of companies with event day losses in excess of 30%, the results of which are available on request. The dendrogram shows two clusters in the data, one of which is overwhelmingly composed of stocks with larger capitalization, higher trading volume and positive trading strategy returns. The other cluster contains predominantly stocks with returns equal to or less than zero, lower capitalization, and lower trading volume. We are able to reject the equality of the means of these variables between the two clusters at the 0.01 level.

V. Conclusion

Various studies have analyzed the phenomenon of price reversals in different time frames and across different markets. Several features distinguish this analysis from those of prior research. This paper broadens the literature in a number of directions. It examines reversals in a new time frame: overnight and intraday return performance is analyzed for stocks with daily close-to-close losses in excess of 10%. The pivotal question the paper addresses is whether contrarian trading strategies based on short-term price reversals have *economically* significant returns. We use a methodology specifically designed to evaluate the economic returns directly by basing the trading strategy return measure on intraday posted quotes. Thus, we are able to gauge realistic returns that a trader following such strategies can attain. The paper also contributes a comprehensive empirical analysis of the existing theories that suggest price reversal explanations based on overreaction, liquidity, and public information flow.

The majority of reversal studies state at one point or another that investors overreact to new information, although there appears to be little empirical evidence relating the reversal phenomenon to the flow of public information. We obtain a new relatively comprehensive measure of the arrival of new information for our events and conduct an explicit test of the relevance of firm specific news releases.

We show evidence in favor of the overreaction hypothesis in the sense that trading returns of reversal-based strategies increase in the absolute value of event day loss. Consistent with behavioral models of Daniel, Hirshleifer, and Subrahmanyam (1998) and Hong and Stein (1999), this study finds reversals to be larger for events unaccompanied by public news releases.

The results are generally supportive of the liquidity pressure hypothesis to the extent that strategy returns increase in event day trading volume. On the other hand, we find that reversals

also increase in company capitalization. Explaining the somewhat puzzling positive relation between reversals and firm size may be a fruitful avenue for future theoretical research.

Prior studies of reversals find that while the average magnitude of price reversals is usually relatively small and does not exceed the average transactions costs, their cross-sectional variability tends to be quite high. Therefore, if variation around the mean is a function of theoretically motivated characteristics, it is possible that profitable trading strategies can be identified based on subsets of event firms.

Guided by the results of the cross-sectional analysis, we are able to identify simple trading rules with economically significant positive returns. A strategy based on event stocks with capitalization and trading volume above the respective 75th percentiles and with relative losses in excess of 30% yields average overnight returns of 1.10%.

In this study we do not carry out conventional adjustments for risk. Although it is unlikely that the magnitudes of trading returns we find can be explained as a compensation for risk, it remains for future research to analyze this question rigorously.

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TABLE 1. Descriptive Statistics

Variable	Mean	Median	Std Dev	Minimum	Maximum	N
Price	24.17	15.32	26.46	5.00	541.83	33,284
Loss	-0.14	-0.13	0.05	-0.79	-0.10	33,284
Market Return	-0.01	-0.01	0.02	-0.07	0.05	33,284
Relative Loss	-0.13	-0.12	0.06	-0.79	-0.03	33,284
News	0.24	0.00	0.43	0.00	1.00	31,076
N_News	0.95	0.00	3.31	0.00	69.00	31,076
Option	0.61	1.00	0.49	0.00	1.00	33,284
Trades	2895	633	8,766	1.00	364,426	28,508
TradeSize	15,171	10,218	54,474	550	6,571,284	28,508
RelTradeSize	0.84	0.64	2.31	0.00	365.88	27,646
Volume	2,097,880	436,321	7,192,954	20.00	318,761,000	33,257
Capitalization (000's)	2,274,871	486,921	10,481,900	113.13	556,962,000	33,277

Note: The sample is composed of stocks with daily close-to-close losses in excess of 10% for the years 2000 - 2001; cross-sectional data are obtained from CRSP, intraday data are obtained from the NYSE TAQ, options data are from CBOE. *Price* is the average closing price on the event day; *Loss* is the close-to-close return on the event day; *Market Return* is the close-to-close return on a value-weighted CRSP portfolio, *Relative Loss* is the difference between the preceding two variables. *News* is a dummy variable equal to one if we are able to locate a news release for the firm from the closing hour of the preceding trading day through the closing hour of the event day; *N_News* is the number of such news releases; *Option* is a dummy variable equal to one if the stock had an option listed on it as of January 1, 2000 or January 1, 2001 for the events in years 2000 and 2001, respectively. *Trades* and *Volume* are for the event day, *Capitalization* numbers are from the preceding day. *TradeSize* is the average dollar value of an event day trade; *RelTradeSize* is the ratio of the latter to the average trade value over days -250 through -20.

TABLE 2. Effects of Event Day Loss on Trading Strategy Returns

Relative Loss	Mean Return, %	T-Stat	# days	# firm-days
All	-1.50	-12.94	492	25358
>=10%	-1.49	-12.48	492	19016
>=15%	-1.35	-7.90	484	6125
>=20%	-1.05	-4.24	443	2231
>=25%	-0.76	-2.52	358	996
>=30%	-0.77	-1.64	272	533
>=35%	-0.60	-0.91	201	314

Note: This table reports average overnight portfolio returns from a trading strategy whereby stocks that experienced close-to-close losses in excess of 10% in the years 2000 - 2001 (or with losses relative to the CRSP value-weighted index in excess of the indicated level) are bought at the average of the ask quotes posted within the last 15 minutes of event day trading and sold at the bids applicable at 9:35 a.m. the next trading day. A portfolio is formed of such stocks each event day with weights determined by the magnitude of the relative losses. Data on capitalization and volume are obtained from CRSP; intraday data are obtained from the NYSE TAQ. *Relative Loss* is the absolute value of the difference between the event day close-to-close loss incurred by the firm and the respective return on the CRSP value-weighted index;

TABLE 3. Logistic Analysis of Information Flow

	Model 1	Model 2	Model 3
Intercept	-2.44 *** (0.00)	-9.32*** (0.00)	-10.90*** (0.00)
RelativeLoss	9.41 *** (0.00)	10.90*** (0.00)	7.07*** (0.00)
LogCap		0.50*** (0.00)	-0.01 0.59
LogVol			0.66*** (0.00)
R ²	0.05	0.13	0.19
N	31076	31070	31050

Note: Our sample is composed of stocks with daily close-to-close losses in excess of 10% for the years 2000 - 2001; cross-sectional data on returns, volume, and firm size are obtained from CRSP. This table summarizes the results of the logistic regressions of the form:

$$News_j = \beta_0 + \beta_1 RelativeLoss_j + \beta_2 LogVol_j + \beta_3 LogCap_j + e_j$$

Where: *News* is a dummy variable equal to one if we are able to locate at least one new release for the firm from the closing hour of the preceding trading day through the closing hour of the event day; *RelativeLoss* is the absolute value of the difference between the event day close-to-close loss incurred by the firm and the respective return on the CRSP value-weighted index; *LogCap* is the natural log of pre-event day capitalization and *LogVol* is the natural log of the event day trading volume; *p*-values are given in parenthesis;

* Significant at the 0.1 level; ** Significant at the 0.05 level; *** Significant at the 0.01 level;

TABLE 4. Cross-Sectional Analysis

	Model 1	Model 2	Model 3	Model 4	Model 5
Intercept	-0.0156*** (0.000)	-0.0888*** (0.000)	-0.0757*** (0.000)	-0.0792*** (0.000)	-0.0852*** (0.000)
Spread	-0.1586*** (0.000)	-0.1419*** (0.000)	-0.0731*** (0.000)	-0.0733*** (0.000)	-0.0729*** (0.000)
RelLoss	0.0316*** (0.000)		0.0121* (0.075)	0.0163** (0.021)	0.019*** (0.008)
LogVol		0.0053*** (0.000)	0.0049*** (0.000)	0.0051*** (0.000)	0.0051*** (0.000)
LogCap		0.0005 (0.126)	0.0013*** (0.002)	0.0013*** (0.002)	0.0012*** (0.006)
News		-0.0018* (0.070)		-0.0031*** (0.001)	-0.0022** (0.035)
N_News		-0.0001 (0.265)			-0.0003* (0.054)
Option					0.0018* (0.062)
TradeSize			-0.0021*** (0.001)	-0.0021*** (0.002)	-0.0023*** (0.001)
Adj. R ²	0.0224	0.0377	0.0230	0.0233	0.0263
N	25013	23334	22559	21191	21191

Note: Our sample is composed of stocks with daily close-to-close losses in excess of 10% for the years 2000 - 2001; cross-sectional data are obtained from CRSP, intraday data are obtained from the NYSE TAQ, options data are from CBOE. This table shows the results of random effects estimations the following model:

$$R_{j,t} = \gamma_0 + \gamma_1 \text{Spread}_{j,t} + \gamma_2 \text{RelLoss}_{j,t} + \gamma_3 \text{LogVol}_{j,t} + \gamma_4 \text{LogCap}_{j,t} + \gamma_5 \text{News}_{j,t} + \gamma_6 \text{N_News}_{j,t} + \gamma_7 \text{Option}_{j,t} + \gamma_8 \text{TradeSize}_{j,t} + v_t + e_{j,t}$$

R - trading returns that can be attained if stocks with close-to-close losses in excess of 10% are bought at the average of ask quotes posted within the last 15 minutes of event day trading and sold at the bid quotes applicable at 9:35 a.m. the next trading day; *Spread* is the difference between the average ask and the average bid quotes posted from 3:45 p.m. to 4 p.m. during the event day trading expressed in percentage terms relative to the midquote point; *RelLoss* is the absolute value of the difference between the event day close-to-close loss incurred by the firm and the respective return on the CRSP value-weighted index; *LogVol* is the natural log of the event day trading volume; *LogCap* is the natural log of the pre-event day capitalization; *News* is a dummy variable equal to one if we are able to locate a new release for the firm from the closing hour of the preceding trading day through the closing hour of the event day; *N_News* is the number of such news releases; *Option* is a dummy variable equal to one if the stock has a CBOE option listed on it and zero otherwise; *TradeSize* is the average dollar size of event day trades; p -values are given in parenthesis; errors are heteroskedastic-consistent.

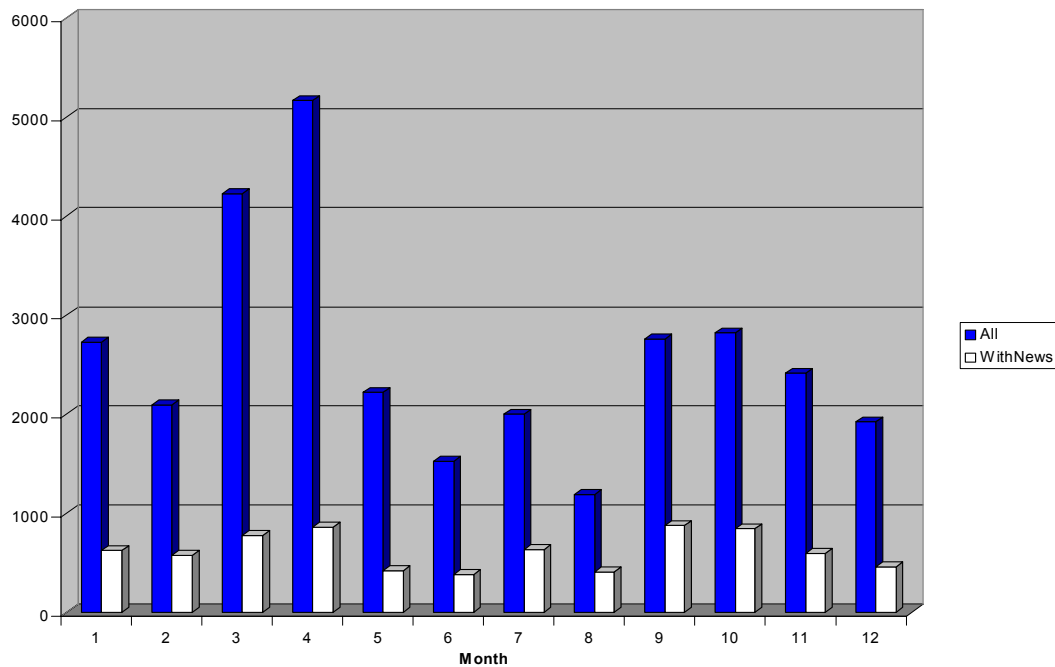
* Significant at the 0.1 level; ** Significant at the 0.05 level; *** Significant at the 0.01 level;

TABLE 5. Trading Strategy Returns: Volume and Firm Size Effects

CAPITALIZATION												
CAPITALIZATION < Q1					CAPITALIZATION >= Q3					TOTAL		
Relative Loss	Mean Return, %	T-Stat	# days	# firm-days	Relative Loss	Mean Return, %	T-Stat	# days	# firm-days	Relative Loss	Mean Return, %	T-Stat
Panel 1					Panel 2					Panel 3		
All	-4.31	-16.78	453	2501	All	-1.56	-3.08	75	117	All	-3.89	-18.96
>=10%	-4.25	-16.23	448	1880	>=10%	-1.47	-2.73	62	82	>=10%	-3.91	-18.67
>=15%	-4.92	-9.19	260	465	>=15%	-0.19	-0.18	21	22	>=15%	-4.61	-11.04
>=20%	-3.45	-3.14	87	110	>=20%	-2.01	-1.71	6	6	>=20%	-4.06	-5.12
>=25%	-4.50	-1.82	28	31	>=25%	-1.42	-2.14	2	2	>=25%	-4.55	-2.41
>=30%	-6.73	-1.62	14	15	>=30%	N/A	N/A	N/A	N/A	>=30%	-6.46	-1.85
>=35%	-8.04	-1.66	7	8	>=35%	N/A	N/A	N/A	N/A	>=35%	-9.09	-2.15
Panel 4					Panel 5					Panel 6		
All	-0.67	-1.28	133	175	All	-0.32	-2.02	451	4852	All	-0.60	-3.90
>=10%	-0.69	-1.28	131	168	>=10%	-0.29	-1.81	449	3518	>=10%	-0.59	-3.78
>=15%	-0.61	-1.21	98	117	>=15%	-0.07	-0.30	347	1147	>=15%	-0.62	-3.00
>=20%	-0.41	-0.75	70	79	>=20%	0.29	0.90	224	426	>=20%	-0.75	-2.74
>=25%	-0.29	-0.50	60	63	>=25%	0.60*	1.48	146	210	>=25%	-0.29	-0.92
>=30%	-0.31	-0.42	42	43	>=30%	1.10***	2.27	87	120	>=30%	-0.45	-0.93
>=35%	-0.10	-0.09	20	20	>=35%	1.73***	2.81	58	73	>=35%	-0.55	-0.77
Panel 7					Panel 8					Panel 9		
All	-2.94	-13.82	484	5002	All	-0.51	-3.48	461	7365	All	-1.50	-12.94
>=10%	-2.88	-13.52	482	3943	>=10%	-0.50	-3.35	459	5221	>=10%	-1.49	-12.48
>=15%	-2.61	-8.09	392	1258	>=15%	-0.21	-0.93	362	1535	>=15%	-1.35	-7.90
>=20%	-1.53	-3.41	246	433	>=20%	0.15	0.47	237	526	>=20%	-1.05	-4.24
>=25%	-1.42	-2.73	149	195	>=25%	0.52*	1.31	151	230	>=25%	-0.76	-2.52
>=30%	-1.68	-2.08	84	96	>=30%	0.96***	1.97	88	127	>=30%	-0.77	-1.64
>=35%	-1.82	-1.73	42	46	>=35%	1.73***	2.81	58	73	>=35%	-0.60	-0.91
Panel 9					Panel 10					Panel 11		
All	-2.94	-13.82	484	5002	All	-0.51	-3.48	461	7365	All	-1.50	-12.94
>=10%	-2.88	-13.52	482	3943	>=10%	-0.50	-3.35	459	5221	>=10%	-1.49	-12.48
>=15%	-2.61	-8.09	392	1258	>=15%	-0.21	-0.93	362	1535	>=15%	-1.35	-7.90
>=20%	-1.53	-3.41	246	433	>=20%	0.15	0.47	237	526	>=20%	-1.05	-4.24
>=25%	-1.42	-2.73	149	195	>=25%	0.52*	1.31	151	230	>=25%	-0.76	-2.52
>=30%	-1.68	-2.08	84	96	>=30%	0.96***	1.97	88	127	>=30%	-0.77	-1.64
>=35%	-1.82	-1.73	42	46	>=35%	1.73***	2.81	58	73	>=35%	-0.60	-0.91

Note: This table reports average overnight portfolio returns from a trading strategy whereby stocks that experienced close-to-close losses in excess of 10% in the years 2000 - 2001 (or with losses relative to the CRSP value-weighted index in excess of the indicated level) are bought at the average of the ask quotes posted within the last 15 minutes of event day trading and sold at the bids applicable at 9:35 a.m. the next trading day. A portfolio is formed of such stocks each event day with weights determined by the magnitude of the relative losses. Data on capitalization and volume are obtained from CRSP; intraday data are obtained from the NYSE TAQ. *Relative Loss* is the absolute value of the difference between the event day close-to-close loss incurred by the firm and the respective return on the CRSP value-weighted index; *Capitalization* is as of the day preceding the event day; *Volume* is the event day trading volume. * Significant at the 0.1 level; ** Significant at the 0.05 level; *** Significant at the 0.01 level; Tests are one-sided.

PANEL 1.



PANEL 2.

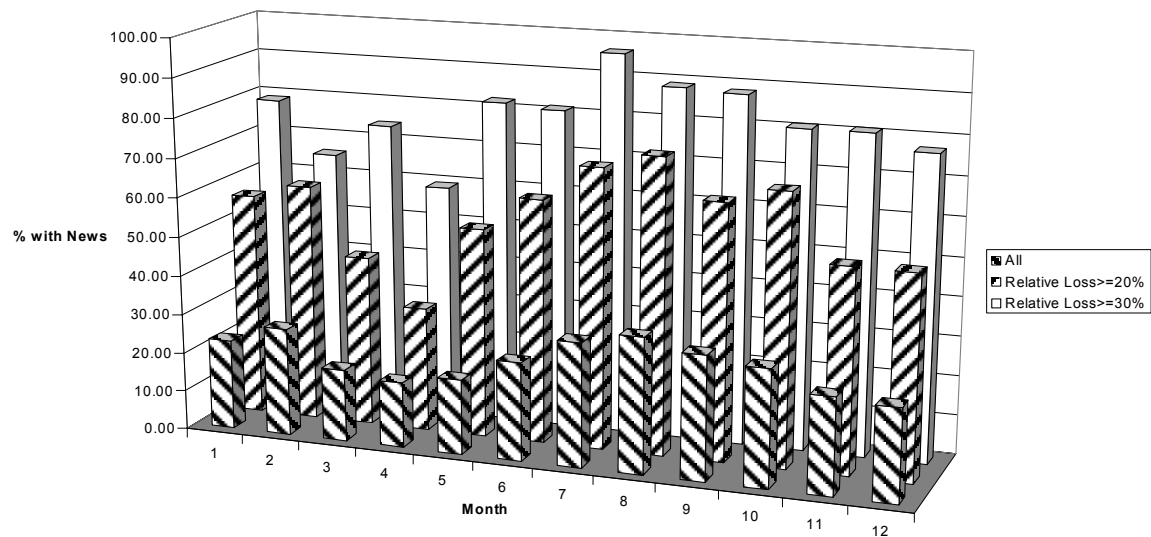
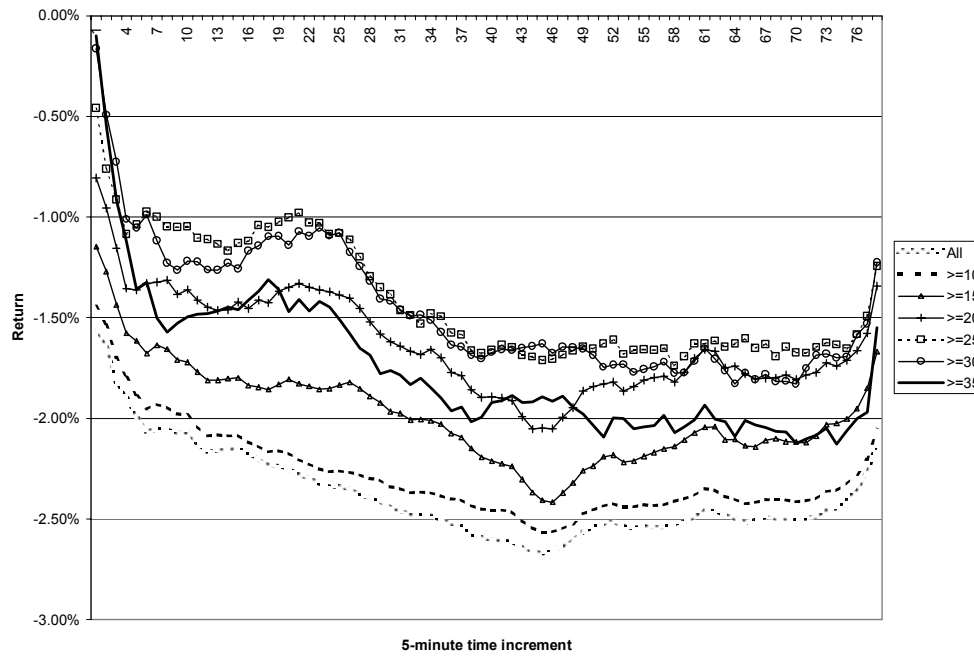


Figure I. Distribution of Sample Events across Months

Note: Panel 1 shows the number of firms whose close-to-close daily losses are in excess of 10% in years 2000 - 2001 by month. *All* is the number of firms whose close-to-close daily losses are in excess of 10%, *With News* is the number of firms for which we are able to locate at least one news release from the closing hour of the trading day preceding the event day through the closing hour of the event day. Panel 2 gives percentages of event firms with news releases by month. *Relative Loss* is equal to the difference between the event day close-to-close return and the equivalent return for CRSP value-weighted index.

PANEL 1.



PANEL 2.

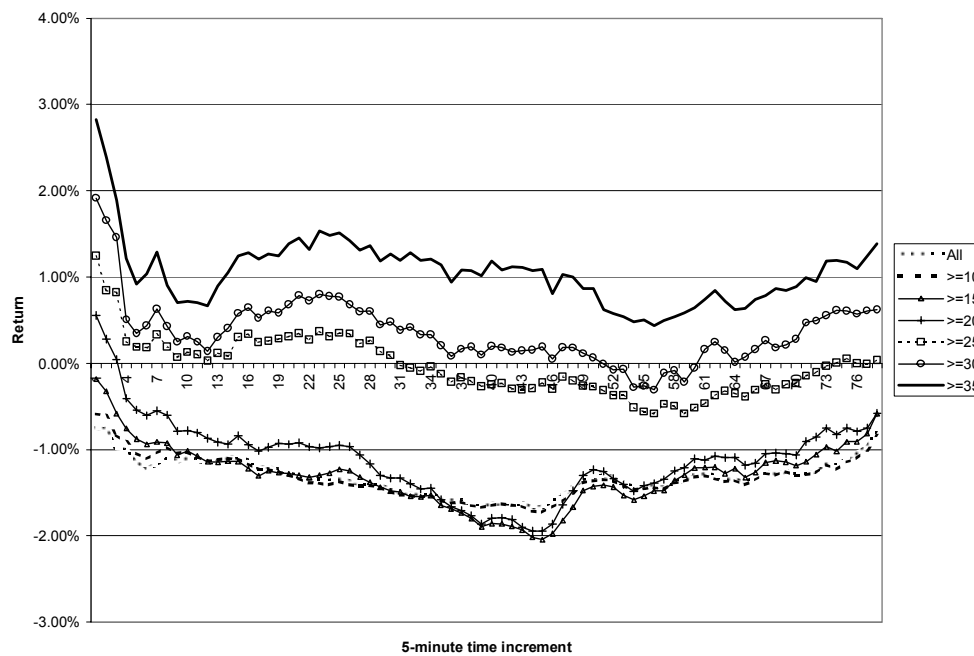
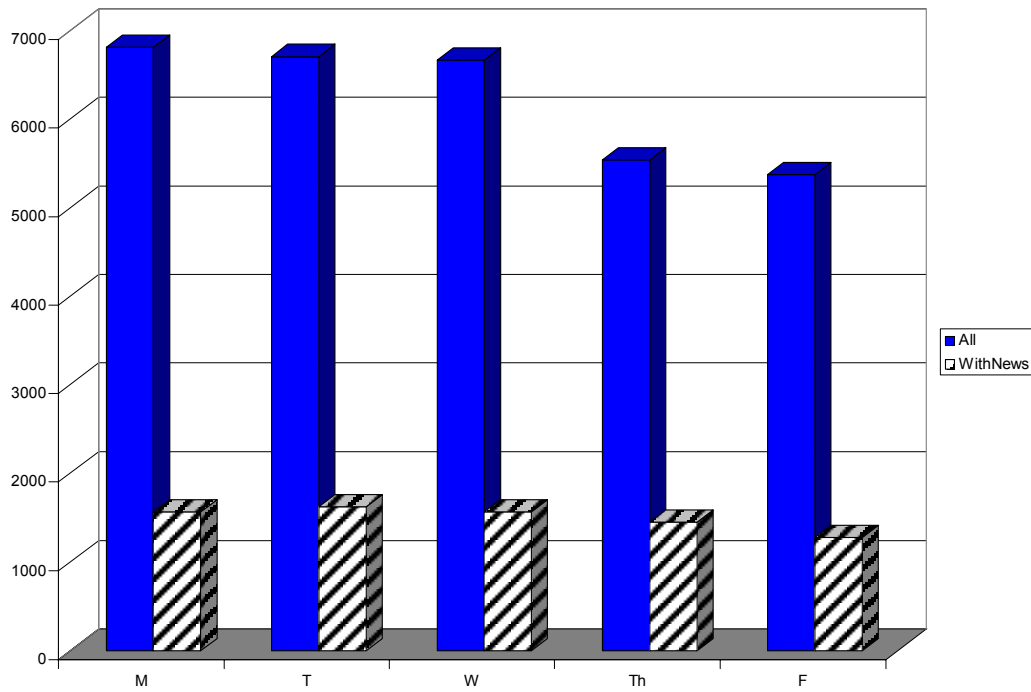


Figure II. Average Trading Strategy Returns by Holding Period

Note: PANEL 1 shows the plot of average returns from a trading strategy whereby stocks with daily close-to-close losses above 10% are bought at the average of ask quotes posted in the last 15 minutes of event day trading and sold at the going bid quote at 78 consecutive five minute increments (9:35 a.m. through 4 p.m.) the next trading day for the overall sample and for subsets with the event day loss relative to the event day market return above the stated level. PANEL 2 depicts a similar plot for companies with capitalization and event day trading volume above the respective 75th percentiles.

PANEL 1.



PANEL 2.

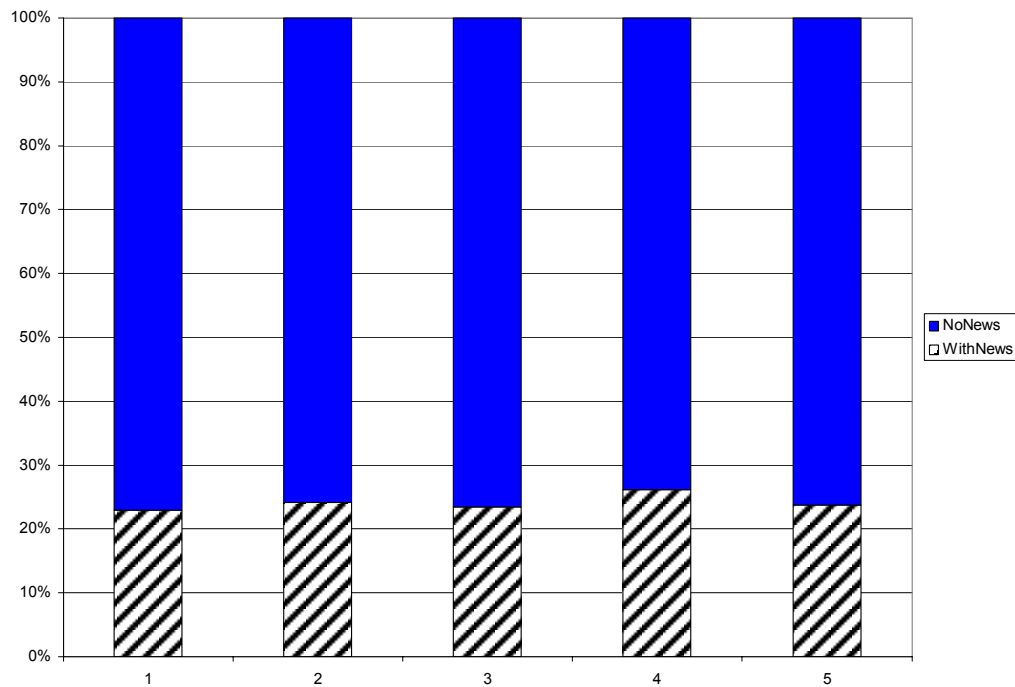
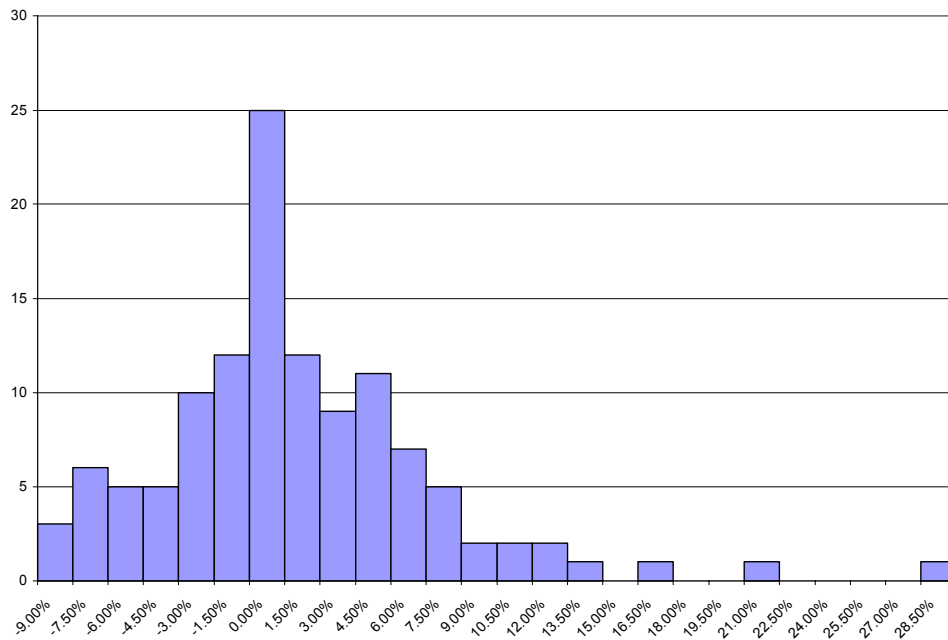


Figure III. Distribution of Sample Events over Weekdays

All is the number of firms whose close-to-close daily losses are in excess of 10%, *With News* is the number of firms for which we are able to locate at least one news release from the closing hour of the trading day preceding the event day through the closing hour of the event day. Both series are plotted across days of week for the years 2000 - 2001.

PANEL 1



PANEL 2.

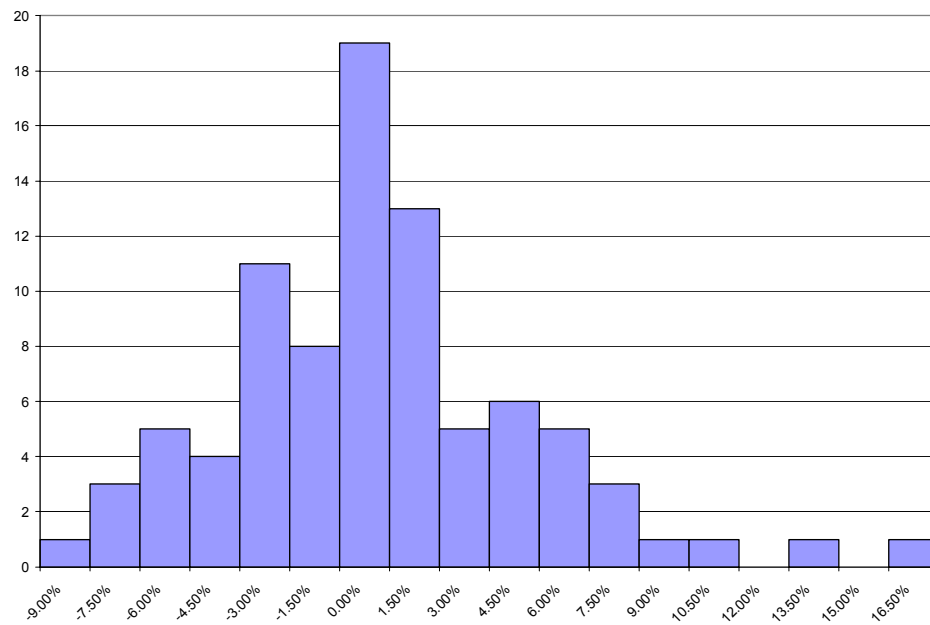


Figure IV. Histograms of Trading Strategy Returns

Note: PANEL 1 shows the distribution of trading returns for the subset of firms with capitalization and trading volume above 75th percentiles and relative event day loss in excess of 30% with each event firm treated separately. PANEL 2 presents the profit distribution for the same subset for a strategy in which same-day firms are combined into a portfolio with weights determined by the relative loss magnitude.

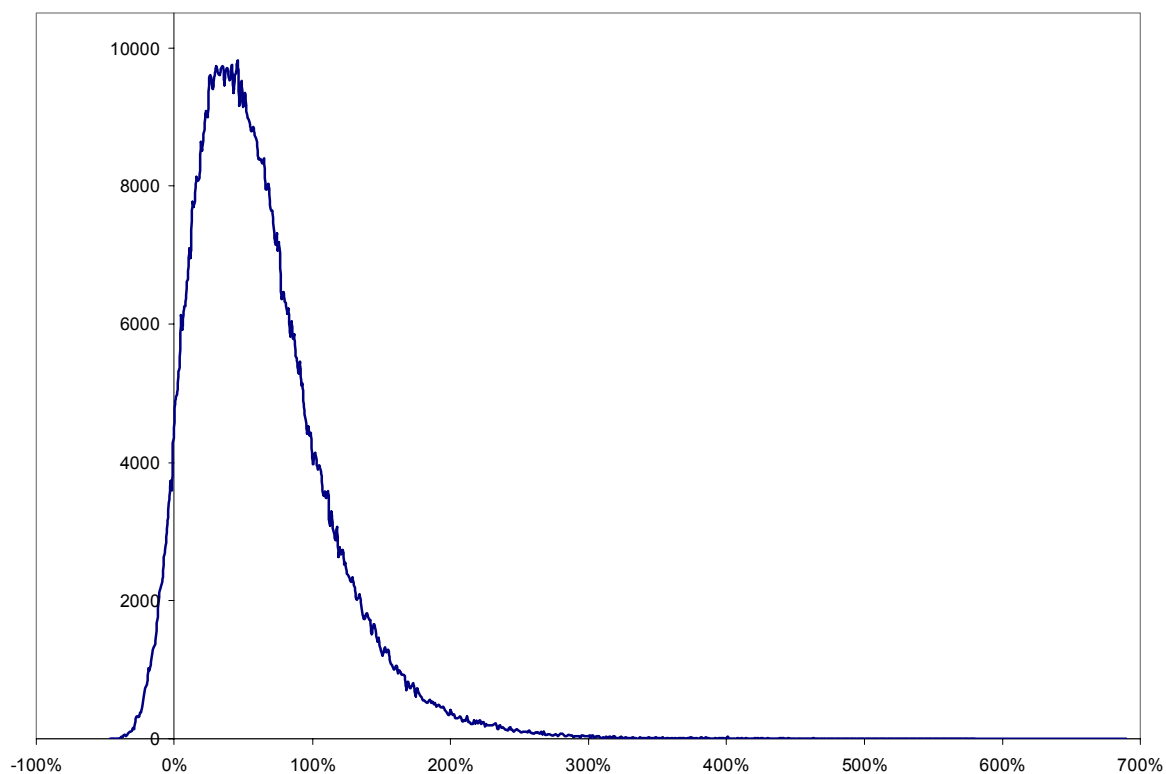


Figure V. Bootstrap Simulation

Note: One million bootstrapped annual trading returns are obtained by sampling with replacement from the return distribution of the strategy in which same-day firms are combined into portfolios with weights determined by the relative loss magnitude. The sample return distribution of the subset of firms with capitalization and trading volume above the 75th percentile and relative event day loss in excess of 30% is used.